Akshat Pandey

Data Platform Engineer

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Summary

Data platform engineer for computer-vision and video-analytics products — I build the streaming **ingest**, the **async processing pipelines**, and the **PostgreSQL/Redis** layers production AI workloads run on: live RTSP/WebRTC ingest via **MediaMTX**, **Celery/RabbitMQ** pipelines that keep inference off the request path, and an internal **MLOps** loop from dataset to retrained model. KAVACH'23 national hackathon winner; B.Tech CS, CPI 9.68.

Technical Skills

Languages: Python, SQL, Bash, TypeScript

Data & Storage: PostgreSQL (JSONB, GIN indexes), Redis, S3, SQLAlchemy 2.x, asyncpg, Alembic

Pipelines & Streaming: Celery, RabbitMQ, ARQ, AWS SQS, asyncio, MediaMTX (RTSP/WebRTC), ONVIF, WebSockets, SSE

Backend & ML: FastAPI, Django, REST, React, MLOps pipelines, scikit-learn, NetworkX, scapy

Cloud & Infra: AWS (Lambda, SQS, S3), Docker, Kubernetes, Helm, Nginx, Linux, Git, Ansible

Professional Experience

Backend Engineer — Data Platform June 2024 – Present
Intozi Tech. Pvt. Ltd. *Gurugram, India*

- Architect the platform's **data and async layers — PostgreSQL**, **Redis**, **RabbitMQ**, **Celery** — so video-processing and model-inference jobs run off the request path and the API stays responsive under load.
- Build the **live-video ingest** for a client-facing **Video Management System**: **Django** services with **MediaMTX** wired in for RTSP/WebRTC, streaming camera feeds and AI analytics into the app in real time.
- Built an internal **MLOps pipeline** (Django + React) that runs the full data loop — dataset upload, auto-labeling, human verification, and model training/retraining — so new models ship without ever leaving the tool.
- Own the **Python** backend across the company's computer-vision and video-analytics products — including the core analytics product **Ikshana** — coordinating delivery within a small team.

Projects

onveef | *Python, ONVIF, httpx, sans-IO* 2026

- Built a **zeep-free ONVIF client library** — the ingest layer that discovers IP cameras, pulls stream URLs, and drives PTZ, imaging and events. Where mainstream Python ONVIF libraries parse WSDL/XSD at runtime, onveef hand-builds SOAP envelopes over a **sans-IO** core: instant imports, and verifiable from recorded fixtures with no hardware.
- Covers ONVIF **Profile S/T/G/M/A/C** and WS-Discovery on two runtime dependencies, with XXE-safe XML parsing throughout.

Papyrus | *FastAPI, Celery, Redis, PostgreSQL, S3, Docker* 2026

- Built a self-hostable document-processing platform on an **asynchronous job pipeline**: a FastAPI API hands merge, split, compress, OCR, reorder and rotate work to **Celery** + **Redis** workers backed by **S3**-compatible object storage, keeping heavy jobs off the request path.
- Designed the async data layer (**SQLAlchemy 2.x**, asyncpg, Alembic) with a **zero-retention mode** that purges documents on a TTL and never logs document content; containerized the full stack with Docker Compose and Kubernetes/Helm deploy artifacts.

MeshHawk | *FastAPI, ARQ, scapy, NetworkX, PostgreSQL, Redis* 2023 – 2026

- Built an **ingest-to-insight pipeline** for **.pcap/.pcapng** captures: scapy reconstructs the 802.11 topology into a directed MAC graph in **NetworkX**, components are scored for 'mesh-ness' by node-degree heuristics, and results land in **PostgreSQL** as **GIN-indexed JSONB** — all local-first, no data leaves the machine.
- Ran the heavy parsing on an **ARQ**/Redis worker behind an async **FastAPI** API, with live monitor-mode capture as a second ingest path and on-demand printable PDF reports rendering each topology as inline SVG.

Also: HeadTogether (20 ORM tables behind 73 REST routes, WebSockets and Redis pub/sub), FOSSLove (async SQLAlchemy, PostgreSQL), Hoctor (scikit-learn), fastapi-boilerplate

Education

Bhilai Institute of Technology Durg, India
Bachelor of Technology in Computer Science – CPI: 9.68 *Aug. 2020 – May 2024*

Achievements

KAVACH'23: Winner of the inaugural nationwide cybersecurity hackathon run by the Government of India — carried on into ShieldBuntu (Ubuntu CIS hardening: 16 idempotent Ansible roles, FastAPI daemon, live SSE).